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Question 1

Not answered

Marked out of 1.00

Question ID #10172: A psychologist develops a diagnostic test to identify people who have injection phobia. In this situation, the test's ________ refers to how good the test is at identifying people who have injection phobia from the pool of people who actually have injection phobia.

Select one:

- A. specificity
- B. sensitivity
- C. positive predictive value
- D. negative predictive value

The correct answer is B.

Sensitivity refers to the probability that a test will correctly identify people with the disease from the pool of people with the disease. It is calculated using the following formula: true positives/(true positives + false negatives).

Answer A: Specificity refers to probability that a test will correctly identify people without the disease from the pool of people without the disease. It is calculated with the following formula: true negatives/(true negatives + false positives).

Answer C: The positive predictive value is the probability that a person identified by the test as having the disease actually has the disease. It is calculated with the following formula: true positives/(true positives + false positives).

Answer D: The negative predictive value is the probability that a person identified by the test as not having the disease does not actually have the disease. The following formula is used to calculate the negative predictive value: true negatives/(true negatives + false negatives).

Question **2**

Not answered

Marked out of 1.00

Question ID #10667: Which of the following is NOT an example of a standard score?

Select one:

- A. WAIS IQ score
- B. percentage score
- C. z-score
- D. T-score

The correct answer is B.

A standard score is a norm-referenced score that indicates an examinee's performance in terms of standard deviation units. Percentages are not standard scores.

Answer A: WAIS IQ scores are standard scores. A WAIS IQ of 115 indicates a score that is one standard deviation above the mean of the norm group.

Answer C: A z-score is a standard score. A z-score of 1.0 indicates a raw score that is one standard deviation above the mean.

Answer D: A T-score is also a standard score. A T-score of 40 indicates a score that is one standard deviation below the mean.

Question **3**

Not answered

Marked out of 1.00

Question ID #11780: _________ refers to the extent to which individual test items contribute to the overall purpose of the test.

Select one:

- A. Validity
- B. Reliability
- C. Discrimination
- D. Relevance

The correct answer is D.

Relevance is determined by judging the extent to which each test item assesses the target content or behavior domain and does so at the appropriate ability level.

Answer A: Validity refers to the overall accuracy of the test rather than to the accuracy of individual test items. In other words, validity addresses the question, "Does the test measure what it was designed to measure?"

Answer B: Reliability refers to the degree to which test scores are free from the effects of measurement error.

Answer C: Discrimination refers to the extent to which test items accurately distinguish between examinees having high and low levels of the attribute(s) measured by the test.

Question **4**

Not answered

Marked out of 1.00

Question ID #11782: A first-year college student obtains a score of 150 on her final English exam, a score of 100 on her math exam, a score of 55 on her chemistry exam, and a score of 30 on her history exam. The means and standard deviations for these tests are, respectively, 125 and 20 for the English exam, 90 and 10 for the math exam, 45 and 5 for the chemistry exam, and 30 and 5 for the history exam. Based on this information, you can conclude that the student's test performance was best on which exam?

Select one:

- A. English
- B. Math
- C. Chemistry
- D. History

The correct answer is C.

The raw scores on different tests may be compared by converting the scores to z-scores, which is done by subtracting the mean from the examinee's raw score to calculate a deviation score and then dividing the deviation score by the standard deviation. The student's chemistry score is equivalent to a z-score of +2.0 and is the highest z-score of the answer choices. Therefore, the student obtained the highest score on the chemistry test.

Answer A: The student's English score is equivalent to a z-score of +1.25.

Answer B: The student's math score is equivalent to a z-score of +1.0.

Answer D: The student's history score is equivalent to a z-score of 0.

Question **5**

Not answered

Marked out of 1.00

Question ID #11783: To maximize the ability of a test to discriminate among test takers, a test developer will want to include test items that vary in terms of difficulty. If the test developer wants to add more difficult items to her test, she will include items that have an item difficulty index of:

Select one:

- A. .90
- B. .50
- C. .10
- D. 0

The correct answer is C.

The item difficulty index ranges from 0 (which occurs when no one answers the item correctly) to 1.0 (which occurs when everyone in a sample answers the item correctly). An item difficulty level of .10 indicates a difficult item (only 10% of examinees in the sample answer it correctly) and is the best answer of those given.

Answer A: When an item's difficulty index is .90, this means that it is a very easy item because 90% of examinees in the sample answer it correctly.

Answer B: An item difficulty level of .50 indicates an item of moderate difficulty because 50% of examinees in the sample answer the item correctly.

Answer D: An item difficulty level of 0 means that no examinees answer this item correctly. Although this is a difficult item, it would not be useful for discriminating among examinees. Therefore, items with a difficulty index of 0 would not be included in a test.

Question **6**

Not answered

Marked out of 1.00

Question ID #11784: A final exam is developed to evaluate students' comprehension of information presented in a high school history class. When the exam is administered to three classes of students at the end of the semester, all students obtain failing scores. This suggests that the exam may have poor ________ validity.

Select one:

- A. concurrent
- B. incremental
- C. content
- D. divergent

The correct answer is C.

The first sentence of this question gives you the information you need to identify the correct answer: It states that the purpose of the test is to assess the students' knowledge of the content presented in the history course. If all students do poorly on a test designed to assess their mastery of the course content, one possible reason is that the test questions do not represent that content, which means the test does not have adequate content validity.

Answer A: Concurrent validity (a type of criterion-related validity) refers to the extent to which test scores correlate with scores on an external criterion. In this situation, performance on the test is the measure of performance, and test scores are not being correlated with scores on an external criterion.

Answer B: Incremental validity is associated with criterion-related validity and refers to the increase in decision-making accuracy that results from use of a predictor.

Answer D: A measure has divergent validity when scores on the measure do not correlate with scores on measures of unrelated traits.

Question 7

Not answered

Marked out of 1.00

Question ID #11785: The correction for attenuation formula is used to measure the impact of increasing:

Select one:

- A. a test's reliability on its validity
- B. a test's validity on its reliability
- C. the number of test items on the test's validity
- D. the number of test items on the test's reliability

The correct answer is A.

The correction for attenuation formula is used to estimate the predictor's validity coefficient if the predictor and/or criterion were perfectly reliable. In this case, the correction for attenuation formula would be used to determine the impact of increasing the reliability of a test on the test's validity.

Answer B: Reliability is a precursor to test validity. As such, there is not a formula to determine the impact of increasing the test's validity on its reliability.

Answer C: Increasing the number of items on a test may increase its reliability and validity; however, this is not measured by the correction for attenuation formula.

Answer D: A test developer would use the Spearman-Brown prophecy formula to estimate the effects of increasing or decreasing the number of test items on the test's reliability coefficient.

Question 8

Not answered

Marked out of 1.00

Question ID #11786: When a test has been constructed on the basis of item response theory, an examinee's total test score provides information about his/her:

Select one:

- A. status on a latent trait or ability
- B. predicted performance on an external criterion
- C. performance relative to other examinees included in the standardization sample
- D. current developmental level

The correct answer is A.

Item response theory is an alternative to classical test theory for the development of tests and interpretation of test scores. Scores on tests developed on the basis of item response theory are reported in terms of the examinee's level on the trait or ability measured by the test rather than in terms of a total score.

Answer B: Criterion-referenced interpretation involves interpreting a test score to predict performance on an external criterion.

Answer C: Norm-referenced interpretation involves comparing an examinee's performance to other examinees in a standardization sample.

Answer D: Information on current developmental level might be provided by some, but not all, tests that are developed on the basis of item response theory. Therefore, this is not the best answer of those given.

Question 9

Not answered

Marked out of 1.00

Question ID #11788: The primary advantage in using a percentile rank, z-score, or T-score is that these scores:

Select one:

- A. are easy to interpret because they reference an individual's test performance to an established standard of performance
- B. are easy to interpret because they reference an individual's test performance to the performance of other examinees
- C. are easy to interpret because they make it possible to predict which criterion group an examinee is likely to belong to
- D. normalize the raw score distribution so that parametric tests can be used to analyze test scores

The correct answer is B.

Raw test scores obtained by examinees have limited meaning. They are often transformed into scores that become meaningful and are easier to interpret. The transformed scores listed in this question are all norm-referenced scores that indicate how well an examinee did in comparison to others in the norm group.

Answer A: This describes a criterion-referenced score, not a norm-referenced score.

Answer C: This answer describes another type of criterion-referenced interpretation that is accomplished by using a regression equation or an expectancy table.

Answer D: Although some norm-referenced scores are "normalized" (e.g., normalized z-scores), this is not true for all norm-referenced scores.

Question 10

Not answered

Marked out of 1.00

Question ID #11790: You would use a multitrait-multimethod matrix in order to:

Select one:

- A. compare a test's predictive and concurrent validity
- B. determine if a test has adequate convergent and discriminant validity
- C. identify the common factors underlying a set of related constructs
- D. test hypotheses about the causal relationships among variables

The correct answer is B.

A multitrait-multimethod matrix contains the correlation coefficients between measures that do and do not purport to measure the same trait. When a measure correlates highly with other measures of the same trait, the measure has convergent validity; when it has low correlations with measures of different traits, it has discriminant (divergent) validity. Convergent and discriminant validity are used as evidence of construct validity, and the multitrait-multimethod matrix contains correlation coefficients that provide information about a measure's convergent and discriminant validity.

Answer A: Predictive and concurrent validity are types of criterion-related validity. The multitrait-multimethod matrix is used to assess a measure's construct validity.

Answer C: This answer describes factor analysis, which is used to identify the common factors that underlie a set of tests, test items, or other variables.

Answer D: This answer describes the aims of conducting research or experiments to test hypotheses about causal relationships among variables.

Question **11**

Not answered

Marked out of 1.00

Question ID #11793: An advantage of using the kappa statistic rather than percent agreement when assessing a test's inter-rater reliability is that the former:

Select one:

- A. is easier to calculate
- B. corrects for chance agreement
- C. corrects for small sample size
- D. takes into account the effects of multicollinearity

The correct answer is B.

The kappa statistic (which is also known as Cohen's kappa and the kappa coefficient) provides a more accurate estimate of reliability than percent agreement because its calculation includes removing the effects of chance agreement.

Answer A: The kappa statistic is more difficult to calculate than percent agreement.

Answer C: The kappa statistic is not used to correct for small sample size. There are several ways to correct for small sample size, including increased sampling, adjusting the alpha level, utilizing different analyses, and changing the effect size.

Answer D: The kappa statistic does not take into account the effects of multicollinearity. Multicollinearity occurs when independent variables in a regression model are correlated.

Question **12**

Not answered

Marked out of 1.00

Question ID #11794: Which type of reliability would be most appropriate for estimating the reliability of a multiple-choice speed test?

Select one:

- A. Split-half
- B. Coefficient of concordance
- C. Alternate forms
- D. Coefficient alpha

The correct answer is C.

Speed tests are designed so that all items answered by an examinee are answered correctly and the examinee's total score depends primarily on his/her speed of responding. Because of the nature of these tests, a measure of internal consistency will provide a spuriously high estimate of the test's reliability. Alternate-forms reliability is an appropriate method for establishing the reliability of speed tests.

Answer A: Split-half reliability is a type of internal consistency reliability and is not appropriate for speed tests.

Answer B: The coefficient of concordance is a measure of inter-rater reliability for three or more raters when ratings are ranks. The test described in this question is a multiple-choice test, not a rating scale or other subjectively scored measure that would report scores in terms of ranks. Therefore, inter-rater reliability would not be a concern.

Answer D: Coefficient alpha is a type of internal consistency reliability and is less appropriate for speed tests.

Question **13**

Not answered

Marked out of 1.00

Question ID #11795: When using principal component analysis:

Select one:

- A. the first principal component represents the largest share of the total variance.
- B. the first principal component represents the smallest share of the total variance.
- C. each component represents an equal share of the total variance.
- D. the order of the components is not related to the share of total variance they represent.

The correct answer is A.

A characteristic of principal components analysis is that the components (factors) are extracted so that the first component reflects the greatest amount of variability, the second component the second greatest amount of variability, etc.

Answer B: The first principal component reflects the largest share of the total variance.

Answer C: Each component represents a share of the total variance, and the components are not equal in their share of the total variance.

Answer D: The components are extracted in order of their share of the total variance.

Question **14**

Not answered

Marked out of 1.00

Question ID #11796: Which of the following scores does not "belong with" the other three?

Select one:

- A. Stanine scores
- B. z-scores
- C. Percentile ranks
- D. Percentage scores

The correct answer is D.

The scores listed in answers A, B, and C are norm-referenced scores that permit an examinee's score to be compared to the scores of others who have taken the same test. In contrast, percentage scores are a type of criterion-referenced score that reference an examinee's score to the content of the exam and indicate how much of the content an examinee has mastered.

Answer A: Stanine scores divide a distribution of scores into nine parts and have a mean of 5 and a standard deviation of approximately 2.

Answer B: Z-scores are norm-referenced scores with a mean of 0 and a standard deviation of 1.

Answer C: Percentile ranks express an examinee's raw score in terms of the percentage of examinees in the norm sample who achieved lower scores.

Question 15

Not answered

Marked out of 1.00

Question ID #11797: Stella obtains a score of 50 on a test that has a standard deviation of 10 and a standard error of measurement of 5. The 95% confidence interval for Stella's score is approximately:

Select one:

- A. 45 to 55
- B. 40 to 60
- C. 35 to 65
- D. 30 to 70

The correct answer is B.

The 95% confidence interval for an obtained test score is constructed by multiplying the standard error of measurement by 1.96 and adding and subtracting the result to and from the examinee's obtained score. An interval of 40 to 60 is closest to the 95% confidence interval and is obtained by multiplying the standard error by 2.0 (instead of 1.96) and then adding and subtracting the result (10) to and from Stella's score of 50.

Answer A: This answer reflects a 68% confidence interval. The 68% confidence interval is constructed by adding and subtracting one standard error to and from the examinee's obtained score. Adding and subtracting one standard error (5) to and from Stella's score of 50 results in a confidence interval of 45 to 55.

Answer C: This answer reflects a 99% confidence interval. The 99% confidence interval is constructed by multiplying the standard error of measurement by 2.58 and adding and subtracting the result to and from the examinee's obtained score. An interval of 35 to 65 is closest to the 99% confidence interval and is obtained by multiplying the standard error (5) by 3.0 (instead of 2.58) and then adding and subtracting the result (15) to and from Stella's score of 50.

Answer D: A confidence interval of 30 to 70 would be the result of multiplying the standard error by 4 and then adding and subtracting the result (20) from Stella's score of 50. This is not a meaningful calculation, as the vast majority of confidence intervals are constructed following the 68-95-99 rule.

Question **16**

Not answered

Marked out of 1.00

Question ID #11799: In a normal distribution, which of the following represents the lowest score?

Select one:

- A. Percentile rank of 20
- B. z-score of -1.0
- C. T-score of 25
- D. Wechsler IQ score of 70

The correct answer is C.

To answer this question, you must be familiar with the relationship between percentile ranks, z-scores, T-scores, and IQ scores in a normal distribution. A T-score is a standardized score with a mean of 50 and a standard deviation of 10. Therefore, a T-score of 25 is two and one-half standard deviations below the mean and represents the lowest score of those given in the answers.

Answer A: A percentile rank of 20 is slightly less than one standard deviation below the mean.

Answer B: A z-score of -1.0 is one standard deviation below the mean.

Answer D: Wechsler IQ scores have a mean of 100 and standard deviation of 15. Therefore, an IQ score of 70 is two standard deviations below the mean.

Question **17**

Not answered

Marked out of 1.00

Question ID #11800: A personnel director uses a mechanical aptitude test to hire machine shop workers. Several of the people hired using the test turn out to be less than adequate performers. These individuals are:

Select one:

- A. true positives
- B. true negatives
- C. false positives
- D. false negatives

The correct answer is C.

False positives are individuals who are predicted to perform satisfactorily by the predictor but, in fact, perform poorly on the criterion. In other words, these individuals have been "falsely identified as positives."

Answer A: True positives are individuals who are predicted to perform satisfactorily by the predictor and, in fact, do well on the criterion.

Answer B: True negatives are individuals who are predicted to perform poorly by the predictor and, in fact, perform poorly on the criterion.

Answer D: False negatives are individuals who are predicted to perform poorly by the predictor but, in fact, do well on the criterion.

Question **18**

Not answered

Marked out of 1.00

Question ID #11801: In terms of magnitude, the standard error of measurement can be:

Select one:

- A. no greater than 1.0
- B. no less than 1.0
- C. no greater than the standard deviation of the test scores
- D. no less than the standard deviation of the test scores

The correct answer is C.

The standard error of measurement equals the standard deviation of the test (predictor) scores times the square root of one minus the reliability coefficient. The maximum value for the standard error of measurement is the value of the standard deviation of the test scores. The standard error is equal to the standard deviation when the reliability coefficient is zero.

Answer A: The standard error can be larger than 1.0.

Answer B: The standard error can also be less than 1.0.

Answer D: The standard error of measurement is equal to 0 when the test's reliability coefficient is +1.0. In this situation, if the test's standard deviation is larger than 0 (which it usually is), the standard error will be less than the standard deviation.

Question **19**

Not answered

Marked out of 1.00

Question ID #11803: A 200-item test that has been administered to 100 college students has a normal distribution, a mean of 145, and a standard deviation of 12. When the students' raw scores have been converted to percentile ranks, Alex obtains a percentile rank of 49, while his twin sister Alicia obtains a percentile rank of 90. The teacher realizes that she made a mistake in scoring Alex's and Alicia's tests: Both should have received a raw score that was five points higher. In terms of their percentile ranks, when the teacher adds the five points to Alex's and Alicia's scores, she can expect that:

Select one:

- A. Alicia's percentile rank will increase more than Alex's.
- B. Alex's percentile rank will increase more than Alicia's.
- C. Alicia's and Alex's percentile ranks will increase by the same amount.
- D. Alicia's and Alex's percentile ranks will not change.

The correct answer is B.

A problem with percentile ranks is that, when the raw scores are normally distributed, raw score differences near the center of the distribution are exaggerated when they are converted to percentile ranks, while raw score differences at the extremes are reduced. (A useful mnemonic for remembering this is "more change in the middle.") Because of the above-described phenomenon, Alex's percentile rank will increase more than Alicia's. This makes sense if you think about the normal distribution: Since most of the scores are "piled up" near the center of the distribution, the increase in 5 points in Alex's score will position him above a larger number of examinee's than the 5-point increase in Alicia's score. This difference will be reflected in their percentile ranks.

Answers A, C, and D: These responses provide incorrect information based on percentile rank- Alex's percentile rank will increase more than Alicia's.

Question **20**

Not answered

Marked out of 1.00

Question ID #11805: In a distribution of percentile ranks, the number of examinees receiving percentile ranks between 20 and 30 is:

Select one:

- A. equal to the number of examinees receiving percentile ranks between 50 and 60
- B. greater than the number of examinees receiving percentile ranks between 50 and 60
- C. about one-half the number of examinees receiving percentile ranks between 50 and 60
- D. about one-fourth the number of examinees receiving percentile ranks between 50 and 60

The correct answer is A.

The flatness of a percentile rank distribution indicates that scores are evenly distributed throughout the full range of the distribution. In other words, at least theoretically, the same number of examinees fall at each percentile rank. Consequently, the same number of examinees obtain percentile ranks between the ranks of 20 and 30, 30 and 40, etc.

Answers B, C, and D: These responses provide incorrect information - the same number of examinees obtain percentile ranks between the ranks of 20 and 30, 30 and 40, 50, and 60, etc.

Question **21**

Not answered

Marked out of 1.00

Question ID #11808: To obtain a "coefficient of stability," you would:

Select one:

- A. administer the same test twice to the same group of examinees on two separate occasions and correlate the two sets of scores.
- B. administer a test to a group of examinees and determine the average inter-item correlation.
- C. administer a test to two different random samples of examinees on two occasions and correlate the two sets of scores.
- D. administer parallel forms of a test to the same group of examinees and correlate the two sets of scores.

The correct answer is A.

The coefficient of stability is another name for the test-retest reliability coefficient. To obtain a coefficient of stability, the same measure is administered to the same group of examinees on two separate occasions and the scores obtained by the examinees are correlated. The result indicates the consistency (stability) of scores over time.

Answer B: This answer describes the procedure for obtaining a coefficient of internal consistency.

Answer C: When assessing test-retest reliability, the test is administered twice to the same group of examinees, not to different groups of examinees.

Answer D: This would provide a coefficient of equivalence (alternate forms reliability coefficient).

Question **22**

Not answered

Marked out of 1.00

Question ID #11811: The best way to control consensual observer drift is to:

Select one:

- A. use the correction for attenuation formula
- B. use a true experimental research design
- C. videotape the observers
- D. alternate raters

The correct answer is D.

Consensual observer drift occurs when observers' ratings become increasingly less accurate over time in a systematic way. Of the actions described in the answers to this question, this one is the best way to alleviate consensual observer drift, which occurs when raters who are working together influence each other's ratings so that they assign ratings in increasingly similar (and idiosyncratic) ways.

Answers A, B, and C: Using the correction for attenuation formula, true experimental research design, or videotaping the observers is not the best way to alleviate consensual observer drift.

Question **23**

Not answered

Marked out of 1.00

Question ID #11813: A reliability coefficient is best defined as a measure of:

Select one:

- A. relevance
- B. consistency
- C. interpretability
- D. generalizability

The correct answer is B.

A reliability coefficient indicates the proportion of variance in test scores that is consistent (i.e., is due to true score variability rather than to measurement error). A test is reliable when it provides consistent results, with inconsistency in test scores being the result of random factors that are present at the time of testing.

Answer A: Relevance refers to the extent to which test items contribute to achieving the stated goals of testing.

Answer C: Interpretability refers to the extent to which the test scores have meaning.

Answer D: Generalizability refers to the extent to which the scores from a sample group apply to a larger population.

Question **24**

Not answered

Marked out of 1.00

Question ID #11814: A test designed to measure knowledge of clinical psychology is likely to have the highest reliability coefficient when:

Select one:

- A. the test consists of 30 items and the tryout sample consisted of individuals who are heterogeneous in terms of knowledge of clinical psychology
- B. the test consists of 30 items and the tryout sample consisted of individuals who are homogeneous in terms of knowledge of clinical psychology
- C. the test consists of 80 items and the tryout sample consisted of individuals who are heterogeneous in terms of knowledge of clinical psychology
- D. the test consists of 80 items and the tryout sample consisted of individuals who are homogeneous in terms of knowledge of clinical psychology

The correct answer is C.

All other things being equal, longer tests are more reliable than shorter tests. In addition, the reliability coefficient (like any other correlation coefficient) is larger when there is an unrestricted range of scores - i.e. when the tryout sample contains examinees who are heterogeneous with regard to the attribute measured by the test.

Answers A, B, and D: These responses provide incorrect information - the highest reliability coefficient is most likely to occur with higher items and the tryout sample consists of individuals who are heterogeneous in terms of knowledge.

Question 25

Not answered

Marked out of 1.00

Question ID #11815: Assuming no constraints in terms of time, money, or other resources, the best way to demonstrate that a test has adequate reliability is by using which of the following techniques?

Select one:

- A. equivalent (alternate) forms
- B. test-retest
- C. Cronbach's alpha
- D. Cohen's kappa

The correct answer is A.

The most thorough method for assessing reliability is the one that takes into account the greatest number of potential sources of measurement error. Because equivalent forms reliability takes into account error due to both time and content sampling, it is the most rigorous method for establishing reliability and, consequently, is considered by some experts to be the best method. Alternate form reliability is often not assessed due to the difficulty in developing forms that are truly equivalent.

Answer B: The primary sources of measurement error for test-retest reliability are any random factors related to the time that passes between the two administrations of the test. Thus, test-retest reliability is not as robust as alternate forms reliability.

Answer C: Cronbach's alpha is another name for coefficient alpha and is used to assess internal consistency reliability.

Answer D: Cohen's kappa is used to measure inter-rater reliability and corrects for chance agreement.

Question **26**

Not answered

Marked out of 1.00

Question ID #11816: In a normal distribution, a T-score of ___ is equivalent to a percentile rank of 16.

Select one:

- A. 10
- B. 20
- C. 30
- D. 40

The correct answer is D.

To identify the correct answer to this question, you need to be familiar with the areas under the normal curve and know that T-scores have a mean of 50 and a standard deviation of 10. In a normal distribution, a T-score of 40 and a percentile rank of 16 are both one standard deviation below the mean.

Answer A: In a normal distribution, a T-score of 10 is four standard deviations below the mean and there is not a meaningful percentile rank that is four standard deviations below the mean.

Answer B: In a normal distribution, a T-score of 20 is three standard deviations below the mean and there is not a meaningful percentile rank that is three standard deviations below the mean.

Answer C: In a normal distribution, a T-score of 30 and a percentile rank of 2.5 are both two standard deviations below the mean.

Question **27**

Not answered

Marked out of 1.00

Question ID #11817: In factor analysis, a factor loading indicates the correlation between:

Select one:

- A. a test and an identified factor
- B. two different tests
- C. two factors measured by the same test
- D. two factors measured by different tests

The correct answer is A.

In factor analysis, a factor loading is a correlation coefficient that indicates the correlation between a test and an identified factor. A factor loading provides information about a test's factorial validity.

Answers B, C, and D: These responses provide incorrect information - in factor analysis, a factor loading indicates the correlation between a test and an identified factor.

Question **28**

Not answered

Marked out of 1.00

Question ID #11818: In factor analysis, when two factors are "orthogonal," this means that:

Select one:

- A. the factors are correlated.
- B. the factors are uncorrelated.
- C. the factors explain a statistically significant amount of variability in test scores.
- D. the factors do not explain a statistically significant amount of variability in test scores.

The correct answer is B.

In factor analysis, orthogonal factors are uncorrelated. This means that the attribute measured by one factor is independent from any attributes measured by the other factor.

Answer A: In factor analysis, oblique factors are correlated because the attributes measured by the factors are not independent.

Answer C: Communality is the amount of true score variability that has been explained by the factor analysis.

Answer D: Specificity is the amount of true score variability that has not been explained by the factor analysis.

Question **29**

Not answered

Marked out of 1.00

Question ID #11819: Which of the following types of validity would you be most interested in when designing a selection test that will be used to predict the future job performance ratings of job applicants?

Select one:

- A. Discriminant
- B. Content
- C. Construct
- D. Criterion-related

The correct answer is D.

When a test is being used to predict performance on a criterion, you would be most interested in the test's criterion-related validity (e.g., in its correlation with the criterion measure). In this situation, you are developing a test that will be used to predict future job performance.

Answer A: Discriminant validity refers to the extent that scores on a test do not correlate with scores on tests measuring different traits. Discriminant validity is of concern for tests designed to measures hypothetical traits (constructs).

Answer B: Content validity is of concern for tests designed to measure a specific content or behavior domain.

Answer C: Construct validity is of concern for tests designed to measure hypothetical traits, such as self-esteem.

Question 30

Not answered

Marked out of 1.00

Question ID #11820: Which of the following best describes the relationship between validity and reliability?

Select one:

- A. A valid test is also a reliable test.
- B. A valid test may or may not be a reliable test.
- C. A reliable test is also a valid test.
- D. An invalid test is not a reliable test.

The correct answer is A.

Knowing that reliability is a necessary but not sufficient condition for validity will help identify the correct answer to this question. Reliability sets an upper limit on validity, which means that a valid test must also be a reliable test. However, high reliability does not guarantee validity - i.e., a test can be free from the effects of measurement error but not measure the attribute it was designed to measure.

Answer B: A valid test must also be reliable.

Answer C: A reliable test may or may not be valid.

Answer D: An invalid test may or may not be reliable.

Question 31

Not answered

Marked out of 1.00

Question ID #11821: In a multitrait-multimethod matrix, the coefficient that indicates a test's reliability is the ____________ coefficient.

Select one:

- A. heterotrait-heteromethod
- B. heterotrait-monomethod
- C. monotrait-heteromethod
- D. monotrait-monomethod

The correct answer is D.

The multitrait-multimethod matrix contains four types of correlation coefficients. The monotrait-monomethod coefficient indicates the correlation of the test with itself and is a measure of the test's reliability.

Answer A: The heterotrait-heteromethod coefficient is a measure of divergent validity.

Answer B: The heterotrait-monomethod coefficient is also a measure of divergent validity.

Answer C: The monotrait-heteromethod coefficient is a measure of convergent validity.

Question **32**

Not answered

Marked out of 1.00

Question ID #11822: Cronbach's alpha is an appropriate method for evaluating reliability when:

Select one:

- A. all test items are designed to measure the same underlying characteristic.
- B. test items are subjectively scored.
- C. the test will be administered to examinees at regular intervals over time.
- D. there is a restriction in the range of scores.

The correct answer is A.

Cronbach's alpha is an appropriate method for evaluating reliability when the test is expected to be internally consistent - i.e., when all test items measure the same or related characteristics. Cronbach's alpha is another name for coefficient alpha and is used to assess internal consistency reliability.

Answer B: Inter-rater reliability is of concern when a test's items are subjectively scored.

Answer C: Test-retest reliability indicates the degree of stability of examinees' scores over time.

Answer D: The reliability coefficient is maximized when a test's range of scores is unrestricted.

Question **33**

Not answered

Marked out of 1.00

Question ID #11823: Incremental validity is a measure of:

Select one:

- A. decision-making accuracy
- B. shrinkage
- C. the generalizability of research results
- D. the costs involved in using a predictor

The correct answer is A.

Incremental validity refers to the increase ("increment") in decision-making accuracy that results from the use of a new predictor (e.g., the increase in accurate hiring decisions).

Answer B: Shrinkage typically occurs during cross-validation. Cross-validation refers to re-assessing a test's criterion-related validity with a new sample. Because the chance factors operating in the original sample are not all present to those operating in the cross-validation sample, the validity coefficient usually "shrinks" for the new sample.

Answer C: External validity typically refers to the generalizability of research results.

Answer D: Incremental validity does not measure the costs involved in using a predictor.

Question **34**

Not answered

Marked out of 1.00

Question ID #11824: In factor analysis, communality refers to:

Select one:

- A. the proportion of variance accounted in a single variable by a single factor.
- B. the proportion of variance accounted in multiple variables by a single factor.
- C. the proportion of variance accounted for in a single variable by all of the identified factors.
- D. the total proportion of variance not explained by the factor analysis.

The correct answer is C.

A communality is a measure of "common variance" and is a reflection of the amount of variance that a test has in common with the other tests included in the factor analysis. In factor analysis, a communality is calculated for each test (or variable) included in the analysis. The communality indicates the total amount of variability accounted for in the test (or variable) by all of the identified factors.

Answer A: This describes a factor loading.

Answer B: This does not describe a communality. A communality indicates the proportion of variance accounted for in a single variable (not multiple variables) by all of identified factors (not a single factor).

Answer D: This describes specificity, which is variability due to factors that are unique to the test and not measured by any other test included in the analysis.

Question **35**

Not answered

Marked out of 1.00

Question ID #11825: Criterion contamination has which of the following effects?

Select one:

- A. It artificially increases scores on the criterion.
- B. It artificially reduces the criterion's reliability coefficient.
- C. It artificially increases the predictor's criterion-related validity coefficient.
- D. It artificially attenuates scores on the predictor and the criterion.

The correct answer is C.

Criterion contamination has the effect of artificially inflating the correlation between the predictor and the criterion. Criterion contamination occurs when a rater's knowledge of a person's predictor performance biases how he/she rates the person on the criterion.

Answer A: The effect of criterion contamination on an individual's criterion score will depend on how well he/she did on the predictor (i.e., if the individual received a low score on the predictor, she will receive a low rating on the criterion and vice versa).

Answer B: The effect on reliability will depend on the situation and the method used to establish the reliability of the criterion.

Answer D: In this context, "attenuate" means to reduce. Since criterion contamination will artificially inflate the predictor's criterion-related validity coefficient, this answer is incorrect.

Question 36

Not answered

Marked out of 1.00

Question ID #11826: To evaluate the concurrent validity of a new selection test for clerical workers, you would:

Select one:

- A. conduct a factor analysis to confirm that the test measures the attributes it was designed to measure
- B. have supervisors and others familiar with the job rate test items for relevance to success as a clerical worker
- C. administer the test to a sample of current clerical workers and correlate their scores on the test with their recently assigned performance ratings
- D. administer the test to clerical workers when they are initially hired and six months after they are hired and then correlate the two sets of scores

The correct answer is C.

Concurrent validity is a type of criterion-related validity. To evaluate a test's criterion-related validity, scores on the predictor (in this case, the selection test) are correlated with scores on a criterion (measure of job performance). When scores on both measures are obtained at about the same time, they provide information on the test's concurrent validity.

Answer A: This technique would be used to evaluate the test's construct validity

Answer B: This would help establish the test's content validity.

Answer D: This procedure would not provide information on the test's concurrent validity.

Question **37**

Not answered

Marked out of 1.00

Question ID #11827: When using criterion-referenced interpretation of scores obtained on a job knowledge test, you would most likely be interested in which of the following?

Select one:

- A. The total number of test items answered correctly by an examinee
- B. An examinee's performance relative to that of other examinees
- C. An examinee's standing on two or more measures designed to assess the same characteristic
- D. Ensuring that test items are based on a systematic job evaluation

The correct answer is A.

As its name implies, criterion-referenced interpretation entails interpreting an examinee's score in terms of a criterion, or standard of performance. One criterion that is used to interpret a person's test score is the total number of correct items. This criterion is probably most associated with "mastery testing." A person is believed to have mastered a content area when he/she obtains a predetermined minimum score on the test that is designed to assess knowledge of that area. There are other types of criteria that are external to the test itself that are used in criterion-referenced interpretation but none of the other responses addresses those types of interpretation; and, consequently, this answer is the best one.

Answer B: A norm-referenced interpretation would entail interpreting an examinee's performance relative to that of other examinees.

Answer C: This answer doesn't describe criterion-referenced interpretation.

Answer D: For some tests used in organizational settings, it might be important to base the content of the test on a job analysis, but not on a job evaluation (which is conducted to set wages and salaries). Also, this is not relevant to criterion-referenced interpretation of test scores.

Question **38**

Not answered

Marked out of 1.00

Question ID #11828: A test developer would use the Kuder-Richardson Formula (KR-20) in order to:

Select one:

- A. evaluate a test's internal consistency reliability
- B. evaluate a test's test-retest reliability
- C. determine the impact of increasing a test's reliability on its validity
- D. determine the impact of lengthening or shortening the test on its reliability

The correct answer is A.

KR-20 is used to determine a test's internal consistency reliability when test items are scored dichotomously.

Answer B: Test-retest reliability is appropriate for determining the reliability of tests designed to measure attributes that are relatively stable over time and that are not affected by repeated measurement.

Answer C: The correction for attenuation formula is used to determine the impact of increasing a test's reliability on its validity.

Answer D: The Spearman-Brown formula is used to determine the impact of lengthening or shortening a test on its reliability.

Question **39**

Not answered

Marked out of 1.00

Question ID #11829: A test developer asks a group of experienced salespeople to review the test items she has developed for a test to help select new sales applicants. This demonstrates the developer is interested in determining the test's ______ validity.

Select one:

- A. incremental
- B. content
- C. concurrent
- D. differential

The correct answer is B.

A test's content validity refers to the extent to which test items represent the domain of knowledge, skills, and/or abilities the test was designed to measure. Content validity is established primarily by having subject matter experts evaluate items in terms of their representativeness.

Answer A: Incremental validity refers to the degree to which the use of a test increases decision-making accuracy.

Answer C: Concurrent validity is a type of criterion-related validity and is evaluated by correlating scores on the test with scores on an external criterion.

Answer D: A test has differential validity when it has different validity coefficients for different groups.

Question **40**

Not answered

Marked out of 1.00

Question ID #11830: To maximize the inter-rater reliability of a behavioral observation scale, you should make sure that coding categories:

Select one:

- A. are mutually exclusive
- B. are measured on an interval or ratio scale
- C. produce criterion-referenced scores
- D. produce scores that are normally distributed

The correct answer is A.

When a person's behavior is to be observed and recorded, that behavior must be operationalized in order for the observations to be meaningful. For example, a psychologist interested in obtaining data about aggressiveness in children might record data using categories such as "hits others" or "destroys property." To maximize the reliability of a behavior observation scale, coding categories must be discrete and mutually exclusive. For example, if the behavioral categories for aggressiveness were "aggressive acts" and "emotional displays," the same behavior might be recorded twice, and an unreliable picture of a child's behavior would be obtained.

Answer B: It is ordinarily better for categories to be discrete rather than continuous (i.e., to be measured on a nominal rather than an interval or ratio scale).

Answer C: This is not a requirement for (or usual characteristic of) behavioral observation scales.

Answer D: Since coding categories are often discrete (nominal), they do not produce scores that are normally distributed.

Question **41**

Not answered

Marked out of 1.00

Question ID #11831: The minimum and maximum values of the standard error of estimate are:

Select one:

- A. -1 and +1
- B. 0 and 1
- C. 0 and the standard deviation of the predictor
- D. 0 and the standard deviation of the criterion

The correct answer is D.

The formula for the standard error of estimate equals the standard deviation of the criterion score times the square root of one minus the validity coefficient squared. This formula indicates that the standard error of estimate ranges from 0 (which occurs when the validity coefficient is 1.0) to the standard deviation of the criterion scores (which occurs when the validity coefficient is 0).

Answers A, B, and C: These responses provide incorrect information - the minimum and maximum values of the standard error of estimate are 0 and the standard deviation of the criterion.

Question **42**

Not answered

Marked out of 1.00

Question ID #11832: When the kappa statistic for a measure is .90, this indicates that the measure:

Select one:

- A. has adequate inter-rater reliability
- B. has adequate internal consistency reliability
- C. has low criterion-related validity
- D. has low incremental validity

The correct answer is A.

Knowing that the kappa statistic (also known as the kappa coefficient) is a measure of inter-rater reliability would have enabled you to identify the correct response to this question. Reliability coefficients range from 0 to +1.0, so a coefficient of .90 indicates good reliability.

Answer B: The kappa statistic is a measure of inter-rater reliability, not internal consistency reliability.

Answer C: The kappa statistic is a measure of reliability, not validity.

Answer D: The kappa statistic is a measure of reliability, not validity.

Question **43**

Not answered

Marked out of 1.00

Question ID #11833: A personnel director hires all job applicants who obtain a high score on a job selection test but, after using the test for six months, realizes that many of the new employees are obtaining low-performance ratings. Assuming that the selection test has adequate criterion-related validity, the personnel director can reduce the number of unsatisfactory workers that she hires using the test by:

Select one:

- A. lowering the selection test cutoff score and the job performance rating cutoff score
- B. raising the selection test cutoff score and the job performance rating cutoff score
- C. lowering selection test cutoff score
- D. raising the selection test cutoff score

The correct answer is D.

Applicants who are hired on the basis of their selection test scores but who perform poorly on the job are false positives. Raising the cutoff score on the selection test (predictor) should reduce the number of individuals who do poorly on the job - i.e., it will reduce the number of positives, including the number of false positives. Note that lowering the job performance rating (criterion) cutoff score would also reduce the number of false positives. However, in many work situations, an employer would not want to do this.

Answers A, B, and C: These responses provide incorrect information - raising the cutoff score on the selection test (predictor) should reduce the number of individuals who do poorly on the job.

Question **44**

Not answered

Marked out of 1.00

Question ID #11837: All other things being equal, which of the following tests is likely to have the largest reliability coefficient?

Select one:

- A. A multiple-choice test that consists of items that each have five answer options
- B. A multiple-choice test that consists of items that each have four answer options
- C. A multiple-choice test that consists of items that each have three answer options
- D. A true-false test

The correct answer is A.

This question is asking about some of the factors that affect the size of the reliability coefficient. All other things being equal, tests containing items that have a low probability of being answered correctly by guessing alone are more reliable than tests containing items that have a high probability of being answered correctly by guessing alone. Of the types of items listed, multiple-choice items with five answer options have the lowest probability of being answered correctly by guessing alone.

Answers B, C, and D: These responses provide the incorrect information - a multiple-choice test that consists of items that each have five answer options is likely to have the largest reliability coefficient.

Question **45**

Not answered

Marked out of 1.00

Question ID #11838: The applicants for sales positions at the Acme Company complain that the selection test they are required to take is unfair because it doesn't "look like" it measures the knowledge and skills that are important for successful job performance. Their complaint suggests that the selection test is lacking which of the following?

Select one:

- A. Incremental validity
- B. Differential validity
- C. Construct validity
- D. Face validity

The correct answer is D.

Face validity refers to the extent that a test appears to be valid to test-takers - i.e., to the extent that the test "looks like" it is measuring what it is supposed to be measuring. In this situation, the selection test doesn't appear to be measuring the skills and knowledge that are important for success as a salesperson.

Answer A: Incremental validity refers to the extent to which a predictor increases decision-making accuracy.

Answer B: A test has differential validity when it has different validity coefficients for different groups.

Answer C: Construct validity is used to determine the usefulness of a test for measuring a theoretical trait or construct.

Question **46**

Not answered

Marked out of 1.00

Question ID #11840: The optimal item difficulty level (p) for a true/false test is:

Select one:

- A. .50
- B. .75
- C. +1.0
- D. -1.0

The correct answer is B.

The optimal difficulty level for test items is halfway between 100% of examinees answering the item correctly and the probability of answering the item correctly by chance alone. For a true/false item, the latter is 50%, so the optimal item difficulty is 75% (.75), which is halfway between 100% and 50%.

Answers A, C, and D: These responses provide incorrect data - the optimal item difficulty level for a true/false test is 75%.

Question **47**

Not answered

Marked out of 1.00

Question ID #11842: According to classical test theory, total variability in test scores is due to:

Select one:

- A. true score variability plus systematic error
- B. true score variability plus random error
- C. relevant variability plus irrelevant variability
- D. relevant variability plus confounding variability

The correct answer is B.

As defined by classical test theory, total variability in test scores is due to a combination of truth and error. This conceptualization is represented by the formula: X = T + E, where X is the total variability in test scores; T is true score variability, and E is variability due to measurement (random) error.

Answers A, C, and D: These responses provide incorrect information - total variability in test scores is due to true score variability plus random error.

Question **48**

Not answered

Marked out of 1.00

Question ID #11843: The assumption underlying convergent validity is that:

Select one:

- A. a measure of a characteristic should correlate highly with a different type of measure that is already known to assess the same characteristic.
- B. a measure of a construct should have little correlation to a measure known to assess an unrelated characteristic.
- C. a measure of a construct should correlate more highly with itself than with another measure of the same construct.
- D. to be valid, a measure of a characteristic should correlate highly with the measure of the behavior it is designed to predict.

The correct answer is A.

A test has convergent validity when it correlates highly with another measure of the same trait. One way to establish a test's construct validity is to determine that it correlates highly with other measures that are already known to assess the same trait. When it does, the measure is said to have convergent validity.

Answer B: This describes discriminant validity (also known as divergent validity).

Answer C: Monotrait-monomethod coefficients are reliability coefficients that indicate the correlation between a measure and itself. They are not directly relevant to a test's convergent and discriminant validity.

Answer D: Predictive validity is a type of criterion-related validity used when the purpose of testing is to predict future status on a criterion.

Question **49**

Not answered

Marked out of 1.00

Question ID #11844: Content sampling is not a potential source of measurement error for which of the following methods for evaluating a test's reliability?

Select one:

- A. Coefficient alpha and alternate forms
- B. Alternate forms and test-retest
- C. Split-half only
- D. Test-retest only

The correct answer is D.

Content sampling refers to the extent to which test scores depend on factors specific to the particular items included in the test (i.e., to its content). Note that this question is asking about the type of reliability that is not affected by content sampling. Because test-retest reliability involves administering the same test (i.e., the same content) twice, content sampling is not a source of error.

Answer A: Coefficient alpha and alternate forms reliability are both subject to content sampling error. Coefficient alpha, a measure of internal consistency, will yield a low reliability coefficient if items are not internally consistent (i.e., if items do not measure the same content domain); alternate forms reliability will be low if the two forms do not assess the same content.

Answer B: Although test-retest reliability is not affected by content sampling error, alternate forms reliability is affected by contact sampling error.

Answer C: Split-half reliability would yield a low reliability coefficient if the two halves of the test do not assess the same content.

Question **50**

Not answered

Marked out of 1.00

Question ID #11845: After reviewing the data collected on a new selection test during the course of a criterion-related validity study on new hires, a psychologist decides to lower the selection test cutoff score. Apparently the psychologist is hoping to do which of the following?

Select one:

- A. Decrease the number of false negatives
- B. Increase the number of true positives
- C. Increase the number of false positives
- D. Increase the number of false negatives

The correct answer is B.

By lowering the selection test (predictor) cutoff score, the psychologist will increase the number of people who are accepted on the basis of their selection test score -- i.e., doing so will increase the number of positives, including the number of true positives, who are individuals who will be selected on the basis of their test scores and will be successful on the criterion.

Answer A: Lowering the predictor cutoff will decrease the number of false negatives. However, it's not likely that the psychologist will want to decrease the number of individuals who would have been successful on the criterion if they had been hired. Therefore, this isn't the best answer.

Answer C: Lowering the selection test cutoff will also increase the number of false positives, since lowering the cutoff score will increase the number of positives overall (both true positives and false positives). The psychologist is apparently hoping to increase the number of true positives by decreasing the cutoff score, but there will also be an increase in the number of false positives. The increase in false positives is not what the psychologist wants; rather, it is an additional outcome of lowering the cutoff.

Answer D: Lowering the selection test cutoff will decrease the number of false negatives.

Question **51**

Not answered

Marked out of 1.00

Question ID #11846: Which of the following item difficulty (p) levels maximizes the differentiation of examinees into high- and low-performing groups?

Select one:

- A. 0.5
- B. 0.9
- C. 1.5
- D. 0.75

The correct answer is A.

An item difficulty level (p) ranges in value from 0 to +1.0 with a value of 0 indicating a very difficult item and a value of +1.0 indicating a very easy item. A difficulty index of .50 indicates that 50% of examinees in the try-out sample answered the item correctly. When p equals .50, this means that the item provides maximum differentiation between the upper- and lower-scoring examinees - i.e., a large proportion of examinees in the upper group answered the item correctly, while a small proportion of examinees in the lower group answered it correctly.

Answers B, C, and D: These responses provide incorrect data- a .50 item difficulty level maximizes the differentiation of examinees into high- and low-performing groups.

Question **52**

Not answered

Marked out of 1.00

Question ID #11847: The standard error of measurement is used to:

Select one:

- A. estimate a test's "true" reliability coefficient
- B. estimate a test's "true" criterion-related validity coefficient
- C. calculate the range within which an examinee's true test score is likely to fall given her obtained score
- D. calculate the range within which an examinee's true criterion score is likely to fall given her predicted criterion score

The correct answer is C.

Since no test is completely reliable, any test score may be in error (i.e., may not reflect the examinee's "true" test score). Consequently, the best strategy is to interpret the examinee's score in terms of a confidence interval. The standard error of measurement is an index of error and is used to construct an interval in which an examinee's true test score is likely to fall given his or her obtained test score.

Answer A: Any test score may be in error, which is why confidence intervals are used instead of a direct estimate of a "true" reliability coefficient.

Answer B: A test's criterion-related validity coefficient can be no greater than the square root of the product of the reliabilities of the predictor and the criterion. Any test score may be made in error, which is why confidence intervals are used instead of a direct estimate of a "true" validity coefficient.

Answer D: This answer describes the use of the standard error of estimate.

Question **53**

Not answered

Marked out of 1.00

Question ID #11849: The distribution of percentile ranks is always:

Select one:

- A. the same as the shape of the distribution of raw scores
- B. normal regardless of the shape of the distribution of raw scores
- C. rectangular (flat) regardless of the shape of the distribution of raw scores
- D. bimodal regardless of the shape of the distribution of raw scores

The correct answer is C.

A distinguishing characteristic of percentile ranks is that their distribution is always rectangular (flat) regardless of the shape of the distribution of raw scores. For the exam, you want to be familiar with the shape of the normal distribution (bell-shaped) and the shape of a distribution of percentile ranks (rectangular).

Answers A, B, and D: these responses provide incorrect information - the distribution of percentile ranks is always rectangular (flat) regardless of the shape of the distribution of raw scores.

Question **54**

Not answered

Marked out of 1.00

Question ID #11850: In factor analysis, the original factor matrix is usually rotated in order to:

Select one:

- A. facilitate interpretation of the identified factors
- B. determine how many factors to extract
- C. cross-validate the factor analysis
- D. verify the causal relationships among the identified factors

The correct answer is A.

One characteristic of the original factor matrix is that it is usually difficult to interpret because it does not provide a clear pattern of factor loadings. The rotation of factors provides a clearer pattern of factor loadings. In the rotated matrix, some tests correlate most highly with one factor, while other tests correlate more highly with a different factor. This makes it easier to identify the factors (dimensions) that account for the intercorrelations between the tests.

Answer B: The correlation matrix provides the information needed to determine how many factors to extract.

Answer C: A confirmatory factor analysis is often used to cross-validate the original factor analysis.

Answer D: Although there is not a technique that can definitively verify the causal relationships among factors, structural equation modeling is often used to analyze the structural relationships between factors.

Question **55**

Not answered

Marked out of 1.00

Question ID #11851: In the multitrait-multimethod matrix, which of the following coefficients provides information about a test's convergent validity?

Select one:

- A. Heterotrait-heteromethod
- B. Heterotrait-monomethod
- C. Monotrait-heteromethod
- D. Monotrait-monomethod

The correct answer is C.

The multitrait-multimethod matrix contains four types of correlation coefficients. The monotrait-heteromethod coefficient is a measure of convergent validity. It indicates the correlation between the test that is being validated and another measure of the same trait (monotrait) that uses a different method of measurement (heteromethod).

Answer A: The heterotrait-heteromethod coefficient is a measure of divergent validity.

Answer B: This heterotrait-monomethod coefficient is also a measure of divergent validity.

Answer D: The monotrait-monomethod coefficient indicates the correlation of the test with itself and is a measure of the test's reliability.

Question **56**

Not answered

Marked out of 1.00

Question ID #11853: When a test user uses a correction for guessing formula that involves subtracting points from each examinee's scores, the resulting distribution of scores will have a __________________ than the original (non-corrected) distribution.

Select one:

- A. higher mean and larger standard deviation
- B. higher mean and smaller standard deviation
- C. lower mean and larger standard deviation
- D. lower mean and smaller standard deviation

The correct answer is C.

The effect of this type of correction for guessing formula on a distribution's mean is fairly easy to understand - i.e., the use of the formula will result in reducing the scores of some examinees and, thereby, reduce the size of the mean. To understand its effect on the standard deviation, assume that the lowest possible score on a test is 0 and that the highest score is 100, which is obtained by at least one examinee. In this situation, as a result of the correction for guessing, some examinees will obtain scores lower than 0, while the highest scorer will still receive a score of 100. When this occurs, the range of scores will increase, and this will be reflected in the distribution's standard deviation.

Answers A, B, and D: These responses provide incorrect information - when a test user uses a correction for guessing formula that involves subtracting points from each examinee's scores, the resulting distribution of scores will have a lower mean and larger standard deviation than the original (non-corrected) distribution.

Question **57**

Not answered

Marked out of 1.00

Question ID #12239: The item discrimination index (D) ranges in value from:

Select one:

- A. 0 to 10
- B. 0 to 50
- C. -1.0 to +1.0
- D. -50 to +50

The correct answer is C.

The item discrimination index indicates the extent to which a test item discriminates between examinees who obtain high versus low scores on the entire test or on an external criterion. The item discrimination index is calculated by subtracting the percent of examinees in the lower scoring group from the percent of examinees in the upper scoring group who answered the item correctly and ranges in value from -1.0 to +1.0.

Answers A, B, and D: These responses provide incorrect data - the item discrimination index ranges in value from -1.0 to +1.0.

Question **58**

Not answered

Marked out of 1.00

Question ID #12241: Which of the following is used to estimate the effects of shortening or lengthening a test on the test's reliability coefficient?

Select one:

- A. Cohen's kappa statistic
- B. Kuder-Richardson Formula 20
- C. Cronbach's coefficient alpha
- D. Spearman-Brown formula

The correct answer is D.

Although the Spearman-Brown formula is probably most often used in conjunction with split-half reliability, it can actually be used whenever a test developer wants to estimate the effects of increasing or decreasing the number of test items on the test's reliability coefficient.

Answer A: Cohen's kappa is used to measure inter-rater reliability and corrects for chance agreement.

Answer B: Kuder-Richardson 20 (KR-20) is a type of internal consistency reliability measure that is used when test items are scored dichotomously.

Answer C: Cronbach's alpha is another name for coefficient alpha and is used to assess internal consistency reliability.

Question **59**

Not answered

Marked out of 1.00

Question ID #12242: A reliability coefficient of .60 indicates that ___ of variability in test scores is true score variability.

Select one:

- A. 60%
- B. 40%
- C. 36%
- D. 16%

The correct answer is A.

A reliability coefficient is interpreted directly as a measure of true score variability. A reliability coefficient of .60 indicates that 60% of variability in scores is true score variability, while the remaining 40% of variability is due to measurement (random) error.

Answers B, C, and D: These responses provide incorrect data - a reliability coefficient of .60 indicates that 60% of variability in test scores is true score variability.

Question **60**

Not answered

Marked out of 1.00

Question ID #12243: In the context of test construction, cross-validation is associated with which of the following?

Select one:

- A. Shrinkage
- B. Criterion deficiency
- C. Criterion contamination
- D. Banding

The correct answer is A.

Cross-validation refers to re-assessing a test's criterion-related validity with a new sample. Because the chance factors operating in the original sample are not all present to those operating in the cross-validation sample, the validity coefficient usually "shrinks" (is smaller) for the new sample.

Answer B: Criterion deficiency refers to the degree to which an actual criterion fails to overlap with the conceptual criterion.

Answer C: Criterion contamination refers to the bias introduced into a person's criterion score as a result of the scorer's knowledge of the person's performance on the predictor.

Answer D: Banding occurs when examinees who obtain scores that fall within a specified range of scores are considered to have received identical scores.

Question **61**

Not answered

Marked out of 1.00

Question ID #12244: A test developer would construct an expectancy table to:

Select one:

- A. facilitate norm-referenced interpretation of test scores
- B. facilitate criterion-referenced interpretation of test scores
- C. correct obtained scores for the effects of guessing
- D. correct obtained test scores for the effects of measurement error

The correct answer is B.

An expectancy table provides the information needed to interpret an examinee's score in terms of expected performance on an external criterion and, consequently, is a method of criterion-referenced interpretation.

Answers A, C, and D: These responses provide incorrect information - a test developer would construct an expectancy table to facilitate criterion-referenced interpretation of test scores.

Question **62**

Not answered

Marked out of 1.00

Question ID #12245: To evaluate the validity of a newly developed selection test for clerical workers, a test developer will correlate scores obtained on the test by newly hired clerical workers with the job performance ratings they receive after being on-the-job for six months. The resulting correlation coefficient will provide information on the test's:

Select one:

- A. discriminant validity
- B. predictive validity
- C. construct validity
- D. concurrent validity

The correct answer is B.

There are two types of criterion-related validity -- predictive and concurrent. As its name implies, predictive validity involves correlating predictor scores with criterion scores that are obtained at a later time to determine how well the predictor predicts future performance on the criterion. The test developer is correlating predictor (selection test) scores with future criterion (job performance) scores and, therefore, is conducting a criterion-related validity study.

Answer A: Discriminant validity is a type of construct validity and involves correlating scores on measures of unrelated characteristics.

Answer C: Construct validity is used to determine the usefulness of a test for measuring a theoretical trait or construct.

Answer D: The test developer would be evaluating concurrent validity (a type of criterion-related validity) if she obtained predictor and criterion scores at about the same time.

Question **63**

Not answered

Marked out of 1.00

Question ID #12246: The point at which an item characteristic curve intercepts the vertical (Y) axis provides information on which of the following?

Select one:

- A. The item's difficulty level
- B. The item's ability to discriminate between low and high scorers
- C. The probability of answering the item correctly by guessing
- D. The item's ability to predict performance on an external criterion

The correct answer is C.

Most item characteristic curves provide information on three parameters -- difficulty level, discrimination, and probability of guessing correctly. The vertical axis indicates the probability of choosing a correct response as a function of an examinee's ability level. The point at which the item characteristic curve intercepts the vertical axis indicates the probability of choosing the correct response by chance alone.

Answer A: In an item characteristic curve, the horizontal axis indicates examinees' ability level and the vertical axis indicates the probability of getting the item correct. Item difficulty is determined by identifying the ability level that is required for there to be a 50% chance of getting the item correct.

Answer B: The slope of the curve indicates the item's ability to discriminate between low and high achievers.

Answer D: The item's ability to predict performance on an external criterion is not indicated on an item characteristic curve.

Question **64**

Not answered

Marked out of 1.00

Question ID #12847: Assuming that each score is from a normal distribution of scores, which of the following accurately lists the scores in order of magnitude from lowest to highest?

Select one:

- A. T = 40; z = 0; PR = 82; WAIS-IV IQ = 126
- B. PR = 16; T = 30; z = 1.5; WAIS-IV IQ = 116
- C. z = -2; T = 25; WAIS-IV IQ = 115; PR = 75
- D. PR = 50; T = 50; WAIS-IV IQ = 132; z = 1.5

The correct answer is A.

For the exam, you want to be familiar with the relationship between norm-referenced scores in a normal distribution. A figure illustrating their relationship is provided in the Test Construction chapter of the written study materials. A T score of 40 is one standard deviation below the mean, a z score of 0 is equal to the mean, a percentile rank (PR) of 82 is slightly below one standard deviation above the mean, and a WAIS-IV score of 126 is slightly below two standard deviations above the mean.

Answers B, C, and D: These responses are not in the correct order.

Question **65**

Not answered

Marked out of 1.00

Question ID #12995: When the heterotrait-monomethod coefficient is large, this indicates:

Select one:

- A. a lack of differential validity
- B. a lack of discriminant validity
- C. adequate convergent validity
- D. adequate concurrent validity

The correct answer is B.

The heterotrait-monomethod coefficient represents the correlation between different traits being measured with the same kind of method. When this coefficient is large, this indicates a lack of discriminant validity. If you are validating a test, you want the heterotrait-monomethod coefficient to be low so that you have evidence of discriminant (divergent) validity.

Answer A: Differential validity refers to differences in validity coefficients across groups. A test that has different validity coefficients for male and female test-takers has differential validity.

Answer C: In a multitrait-multimethod matrix, convergent validity is present and monotrait-heteromethod coefficients are large. This indicates a high correlation between different measures of the same trait.

Answer D: Concurrent validity is a type of criterion-related validity and is of concern when the purpose of the test is to estimate an individual's current status on some external criterion. Concurrent validity is not represented in the multitrait-multimethod matrix.

Question **66**

Not answered

Marked out of 1.00

Question ID #13458: It would be most important to assess the test-retest reliability of a measure that:

Select one:

- A. is subjectively scored
- B. assesses examinees' speed of responding
- C. measures a stable trait
- D. measures a characteristic that fluctuates over time

The correct answer is C.

Test-retest reliability determines the reliability of tests designed to measure traits that are stable over time. To evaluate test-retest reliability, the same test is administered to the same group of examinees on two different occasions. The two sets of scores are then correlated.

Answer A: Test-retest reliability could be used to assess the reliability of a subjectively scored test, but it would not be "important" to do so.

Answer B: Alternate forms reliability is better for speeded tests because it eliminates the problem of practice effects.

Answer D: Internal consistency reliability is useful when a test is designed to measure a characteristic that fluctuates over time.

Question **67**

Not answered

Marked out of 1.00

Question ID #13627: In terms of item response theory, the slope (steepness) of the item characteristic curve (ICC) indicates the item's:

Select one:

- A. level of difficulty
- B. ability to discriminate between examinees
- C. internal consistency reliability
- D. criterion-related validity

The correct answer is B.

The steeper the slope of the item characteristic curve (ICC), the better its ability to discriminate between examinees who are high and low on the characteristic being measured.

Answer A: An item's level of difficulty is represented by an ICC but is indicated by the ability level (charted on the horizontal axis) at which 50% of examinees in the sample answer the item correctly.

Answer C: Reliability refers to a test's consistency and is not indicated by the item characteristic curve.

Answer D: Validity refers to a test's accuracy and is not indicated by the item characteristic curve.